

How Chatbots Really Work: Teacher Guide

A free Hour of AI lab by Laurence Moroney · CS Education Week, Dec 8-14, 2026 · Companion to Your First AI Model in an Hour

Student notebook: https://colab.research.google.com/github/lmoroney/hourofai/blob/main/lab2-how-chatbots-really-work/how_chatbots_really_work.ipynb · **Lab home / downloads:** [HOSTING_PAGE_PLACEHOLDER]

Audience: ages 13-18, no ML background, little or no Python · **Time:** 45-60 min · **Cost:** free, no API keys or accounts beyond Google

Most students meet AI through chatbots. This lab lifts the lid: students load a real (small, open) language model in the browser, watch it predict the next word, change its creativity, see how it's turned into a chatbot, and then catch it making things up. Lab 1 is helpful but not required.

Learning goals

By the end, students can:

1. Explain that a chatbot is, at its core, a **next-word (token) predictor** trained on lots of human text.
2. Describe **tokens** and why models don't "see" letters.
3. Explain how **temperature** and random sampling make answers vary.
4. Describe how a base model becomes a chatbot: **instruction tuning**, **chat templates**, and **system prompts**.
5. Identify and explain **hallucinations**, **sensitivity to wording**, and **bias**, and apply **verification habits** when using chatbots for school.

Setup (5 minutes, before class)

- **No installs, no API keys, no paid services.** Runs on free Google Colab CPU (no GPU needed).
- Part 0 downloads two small open models (**SmolLM2-135M** and **SmolLM2-135M-Instruct** from Hugging Face, ~540 MB total). On Colab this usually takes under a minute. **Tip:** have students run Part 0 as soon as they sit down.
- If a whole class downloads at once on a slow school network, stagger starts or run Part 0 while you introduce the hook.
- Open https://colab.research.google.com/github/lmoroney/hourofai/blob/main/lab2-how-chatbots-really-work/how_chatbots_really_work.ipynb → **File** → **Save a copy in Drive**. Do a dry run: **Runtime** → **Run all** (about 1-3 minutes).
- The model is **tiny** (135M parameters vs hundreds of billions for big chatbots). It's less capable and its mistakes are more obvious. That's intentional, and the notebook says so.

Timing plan

Part	What happens	Min
Hook + setup (Part 0)	"A chatbot is a next-word predictor", like phone autocomplete. Start the model download	5
1. What's an LLM?	Trained by guessing hidden words, billions of times; predicts likely text, not true text	4
2. Tokens		6

Part	What happens	Min
	See text split into tokens and numbers; try names, slang, emoji, "strawberry"	
3. Next word	Guess first, then see the model's probabilities; watch it write token by token	8
4. Temperature	Compare probability charts; poems at 0.2 / 1.0 / 2.0; same question, different answers	8
5. Making a chatbot	Base vs chat model; the hidden chat template; system prompt experiment	8
6. Where chatbots go wrong	Invented scientist, fake sources, contradicting answers, counting/math, he/she bias chart	12
7. Using chatbots wisely	Tips, discussion, exit ticket	5-8

Short on time (45 min)? Skip the goldfish cell in Part 4 and the system-prompt experiment in Part 5. **Never skip Parts 6-7.**

What students should see (from our test run)

- "Once upon a time, there was a" gives: little 12%, curious 10%, wise 10%.
- "The capital of France is" gives "the" (27%) ahead of "Paris" (8%). It predicts likely text, not facts.
- **Invented scientist** "Dr. Maribel Quintero-Zhang": a full fake biography (born 1934 in San Francisco, a Berkeley PhD, quantum mechanics).
- **Fake sources:** invented titles and historians ("Dr. John Smith and Dr. Michael Thompson").
- **Tomato:** the neutral question and the leading question get contradicting answers.
- **Strawberry / 17×23:** wrong answers (the correct answers are 3 and 391).
- **Bias chart:** "he" is more likely for doctor, engineer, CEO, and pilot; "she" for nurse and babysitter.
- Answers with temperature 0 should match the above. Sampled answers (temperature > 0) will vary, which is the point.

Discussion questions

1. How would you explain to a younger kid what a chatbot actually is?
2. The bot invented a scientist and sources and sounded confident. When could that cause real harm (homework, health, news, law)?
3. Why is "I don't know" hard for a next-word predictor?
4. Whose writing is over-represented in training data, and whose is missing? How could that shape answers? (Also: languages that get split into more tokens.)
5. Companies write hidden system prompts. What might they include? Should users see them?
6. What's a good use and a bad use of a chatbot for schoolwork?

Common hiccups and fixes

Symptom	Fix
Part 0 is slow or stuck	It's downloading ~540 MB. Wait; rerun if it errors. Check the firewall allows huggingface.co.
Warning: "unauthenticated requests to the HF Hub"	Harmless. Ignore it. No token is needed.

Symptom	Fix
NameError: ... is not defined	A cell was skipped. Runtime → Run all.
Bot's answer stops mid-sentence	Replies are capped (max_words) to keep things fast. Raise the number if you like.
"My answer is different from my neighbour's"	Expected when temperature > 0 (random sampling).
Bot ignores the system prompt	Tiny models follow instructions poorly. It's a teaching point (see Part 5 note).
Odd or inappropriate text at high temperature	High temperature produces random text. Lower it. Remind students it's a probability machine, not a person.
Runtime disconnected	Runtime → Run all (the models re-download).

Extensions for fast finishers

- Invent a fake person, book, or event and test whether the bot makes up facts. Can any wording get it to say "I don't know"?
- Find prompts where the base and chat models differ the most.
- Add jobs to the bias chart. Try other patterns such as "The ____ was very".
- Build a "neutral vs leading" pair of questions that flips the answer.
- Try Lab 1 (Your First AI Model in an Hour) to train an image classifier: https://colab.research.google.com/github/lmoroney/hourofai/blob/main/lab1-first-model-in-an-hour/first_model_in_an_hour.ipynb

Alignment to AI literacy themes

- **How AI works:** training on data, tokens, prediction, probabilities, sampling (Parts 1-4).
- **Humans shape AI:** instruction tuning, human feedback, system prompts, choice of data (Part 5).
- **Evaluating AI output:** hallucination, inconsistency, confidence vs correctness, verification (Part 6).
- **Bias and societal impact:** stereotypes absorbed from data, uneven support across languages (Parts 2, 6).
- **Responsible use:** academic honesty, privacy, source checking (Part 7).
- Maps well to AI4K12's "Five Big Ideas" (Representation & Reasoning, Learning, Natural Interaction, Societal Impact) and CSTA impacts-of-computing standards.
[ALIGNMENT_DETAILS_PLACEHOLDER: confirm specific standard codes before publishing]

Note on content: output from the small model is unfiltered. It behaved appropriately in testing, but at high temperature or with student-invented prompts it can produce odd text. Preview the notebook before class.

Questions or feedback? [CONTACT_PLACEHOLDER] — Laurence